SE LAB-05

Static Code Analysis - Parking Lot System

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Section: H

Github Link:
https://github.com/smonty-19/PES2UG24CS491_SE_LAB5_Parking_Lot_System.git

Running the initial python file

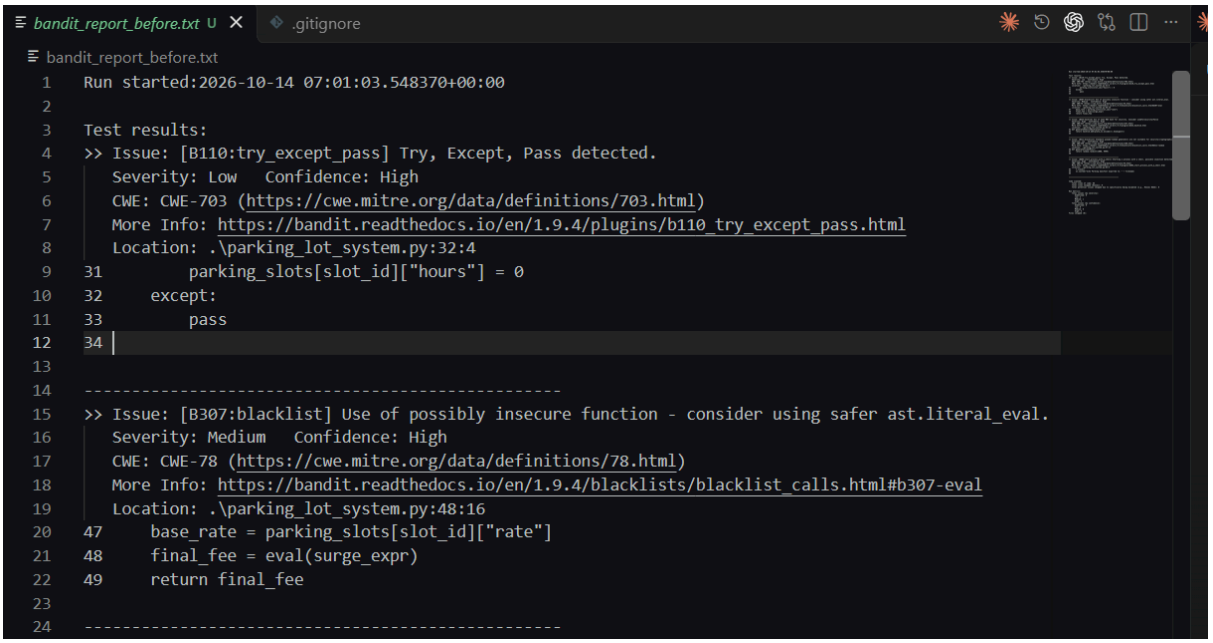
```
(.venv) PS C:\Users\shrey\OneDrive\Documents\SE LAB\Lab5_StaticCodeAnalysis> python parking_lot_system.py
Is P02 available: False
Surge Parking Fee: 45.0
Vehicle Hash: 353ee89fa6aac8992f8fbceb823f602c
Ticket PIN: 5248
Parking manifest exported to: parking_manifest.txt
Vacant slots: 1
Barrier Key active: BARRIER_GATE_SECRET_TOKEN_2026
(.venv) PS C:\Users\shrey\OneDrive\Documents\SE LAB\Lab5_StaticCodeAnalysis>
```

Pylint Report Before Changes

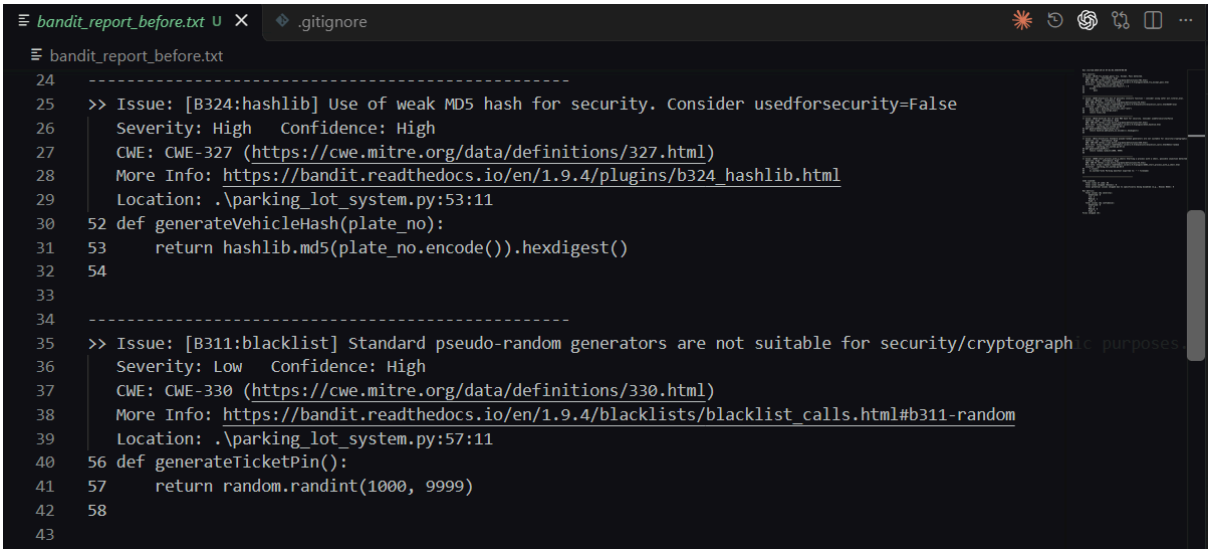
```
************* Module parking_lot_system
parking_lot_system.py:23:0: C0301: Line too long (131/100) (line-too-long)
parking_lot_system.py:88:0: C0304: Final newline missing (missing-final-newline)
parking_lot_system.py:1:0: C0114: Missing module docstring (missing-module-docstring)
parking_lot_system.py:12:0: C0103: Constant name "total_collections" doesn't conform to UPPER_CASE naming style (invalid-name)
parking_lot_system.py:15:0: C0116: Missing function or method docstring (missing-function-docstring)
parking_lot_system.py:15:0: C0103: Function name "parkVehicle" doesn't conform to snake_case naming style (invalid-name)
parking_lot_system.py:15:0: W0102: Dangerous default value [] as argument (dangerous-default-value)
parking_lot_system.py:16:4: W0603: Using the global statement (global-statement)
parking_lot_system.py:23:22: C0209: Formatting a regular string which could be an f-string (consider-using-f-string)
parking_lot_system.py:27:0: C0116: Missing function or method docstring (missing-function-docstring)
parking_lot_system.py:27:0: C0103: Function name "releaseSlot" doesn't conform to snake_case naming style (invalid-name)
parking_lot_system.py:32:4: W0702: No exception type(s) specified (bare-except)
parking_lot_system.py:36:0: C0116: Missing function or method docstring (missing-function-docstring)
parking_lot_system.py:36:0: C0103: Function name "checkSlotAvailability" doesn't conform to snake_case naming style (invalid-name)
parking_lot_system.py:38:7: C0121: Comparison 'parking_slots.get(slot_id) == None' should be 'parking_slots.get(slot_id) is None' (singleton-comparison)
parking_lot_system.py:40:4: R1703: The if statement can be replaced with 'return bool(test)' (simplifiable-if-statement)
parking_lot_system.py:40:4: R1705: Unnecessary "else" after "return", remove the "else" and de-indent the code inside it (no-else-return)
parking_lot_system.py:37:4: W0612: Unused variable 'overtime_fee' (unused-variable)
parking_lot_system.py:46:0: C0116: Missing function or method docstring (missing-function-docstring)
parking_lot_system.py:46:0: C0103: Function name "calculateParkingFee" doesn't conform to snake_case naming style (invalid-name)
parking_lot_system.py:48:16: W0123: Use of eval (eval-used)
parking_lot_system.py:47:4: W0612: Unused variable 'base_rate' (unused-variable)
parking_lot_system.py:52:0: C0116: Missing function or method docstring (missing-function-docstring)
parking_lot_system.py:52:0: C0103: Function name "generateVehicleHash" doesn't conform to snake_case naming style (invalid-name)
parking_lot_system.py:56:0: C0116: Missing function or method docstring (missing-function-docstring)
parking_lot_system.py:56:0: C0103: Function name "generateTicketPin" doesn't conform to snake_case naming style (invalid-name)
parking_lot_system.py:60:0: C0116: Missing function or method docstring (missing-function-docstring)
parking_lot_system.py:60:0: C0103: Function name "exportParkingManifest" doesn't conform to snake_case naming style (invalid-name)
parking_lot_system.py:61:8: W1514: Using open without explicitly specifying an encoding (unspecified-encoding)
parking_lot_system.py:61:8: R1732: Consider using 'with' for resource-allocating operations (consider-using-with)
parking_lot_system.py:67:0: C0116: Missing function or method docstring (missing-function-docstring)
parking_lot_system.py:67:0: C0103: Function name "getAvailableSummary" doesn't conform to snake_case naming style (invalid-name)
parking_lot_system.py:67:24: W0621: Redefining name 'parking_slots' from outer scope (line 6) (redefined-outer-name)
parking_lot_system.py:69:8: W0612: Unused variable 's' (unused-variable)
parking_lot_system.py:75:0: C0116: Missing function or method docstring (missing-function-docstring)
parking_lot_system.py:4:0: W0611: Unused import math (unused-import)

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Your code has been rated at 4.10/10
```

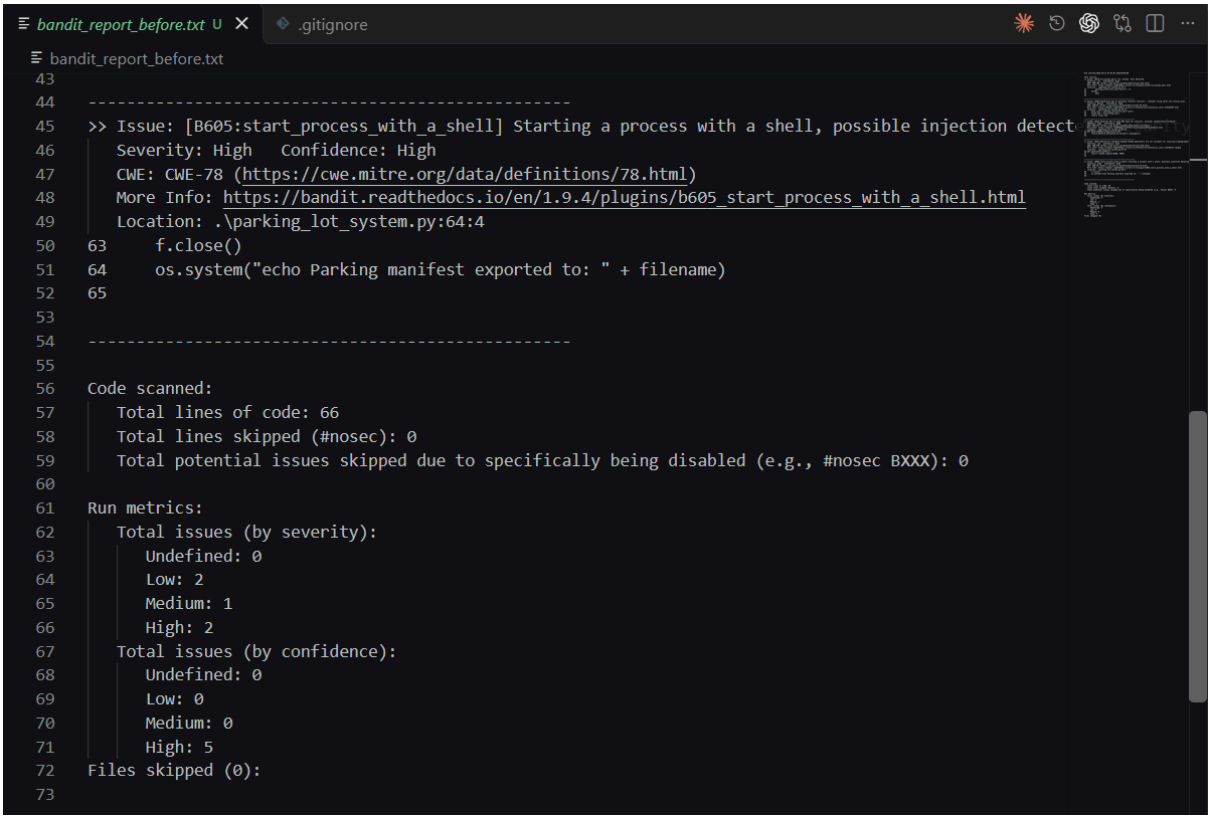
Bandit Report Before Changes



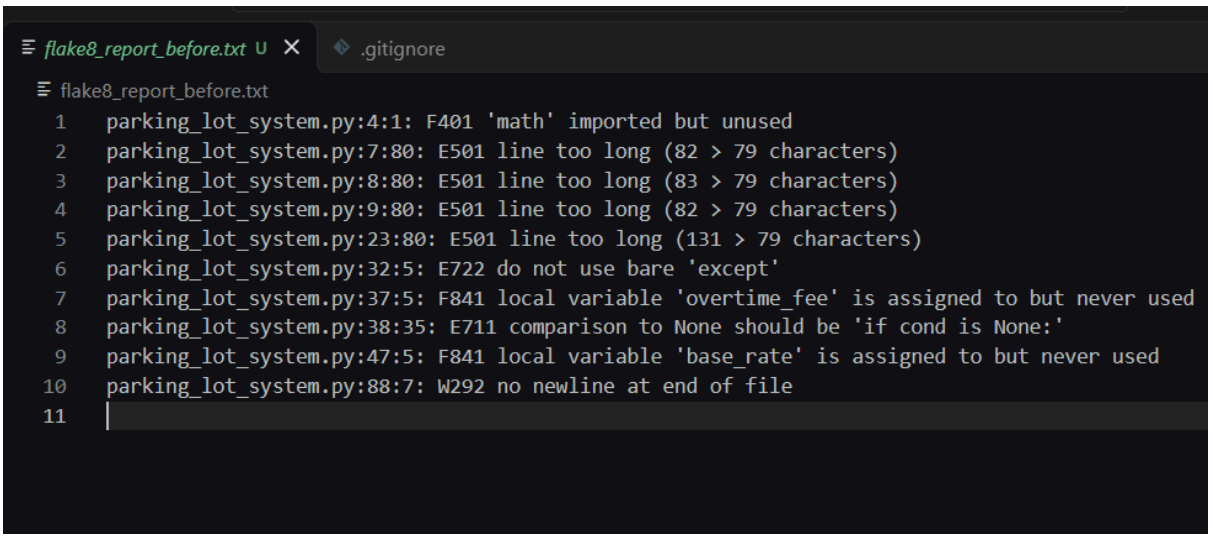
```
bandit_report_before.txt U  .gitignore
bandit_report_before.txt
1  Run started:2026-10-14 07:01:03.548370+00:00
2
3  Test results:
4  >> Issue: [B110:try_except_pass] Try, Except, Pass detected.
5     Severity: Low   Confidence: High
6     CWE: CWE-703 (https://cwe.mitre.org/data/definitions/703.html)
7     More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b110_try_except_pass.html
8     Location: .\parking_lot_system.py:32:4
9  31         parking_slots[slot_id]["hours"] = 0
10 32     except:
11 33         pass
12 34
13
14 --------------------------------------------------
15 >> Issue: [B307:blacklist] Use of possibly insecure function - consider using safer ast.literal_eval.
16    Severity: Medium   Confidence: High
17    CWE: CWE-78 (https://cwe.mitre.org/data/definitions/78.html)
18    More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b307-eval
19    Location: .\parking_lot_system.py:48:16
20 47     base_rate = parking_slots[slot_id]["rate"]
21 48     final_fee = eval(surge_expr)
22 49     return final_fee
23
24 --------------------------------------------------
```



```
bandit_report_before.txt U  .gitignore
bandit_report_before.txt
24 --------------------------------------------------
25 >> Issue: [B324:hashlib] Use of weak MD5 hash for security. Consider usedforsecurity=False
26    Severity: High   Confidence: High
27    CWE: CWE-327 (https://cwe.mitre.org/data/definitions/327.html)
28    More Info: https://bandit.readthedocs.io/en/1.9.4/plugins/b324_hashlib.html
29    Location: .\parking_lot_system.py:53:11
30 52 def generateVehicleHash(plate_no):
31 53     return hashlib.md5(plate_no.encode()).hexdigest()
32 54
33
34 --------------------------------------------------
35 >> Issue: [B311:blacklist] Standard pseudo-random generators are not suitable for security/cryptographic purposes.
36    Severity: Low   Confidence: High
37    CWE: CWE-330 (https://cwe.mitre.org/data/definitions/330.html)
38    More Info: https://bandit.readthedocs.io/en/1.9.4/blacklists/blacklist_calls.html#b311-random
39    Location: .\parking_lot_system.py:57:11
40 56 def generateTicketPin():
41 57     return random.randint(1000, 9999)
42 58
43
```



Flake8 Report Before Changes

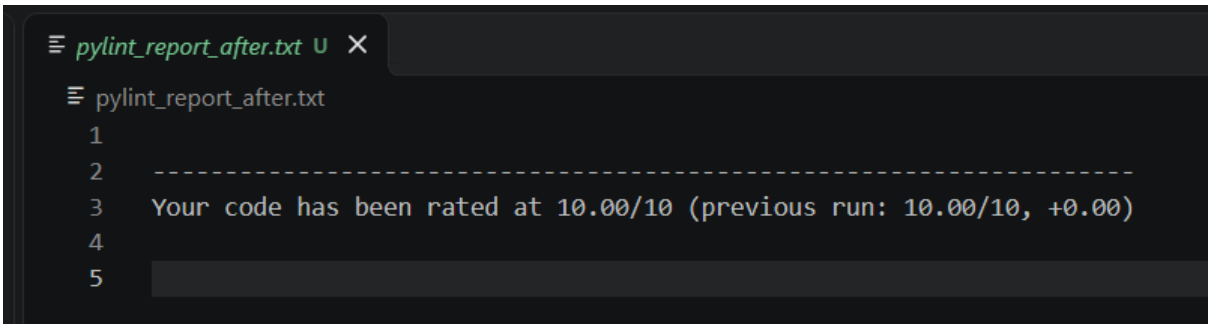


Issues Table with approach used to fix

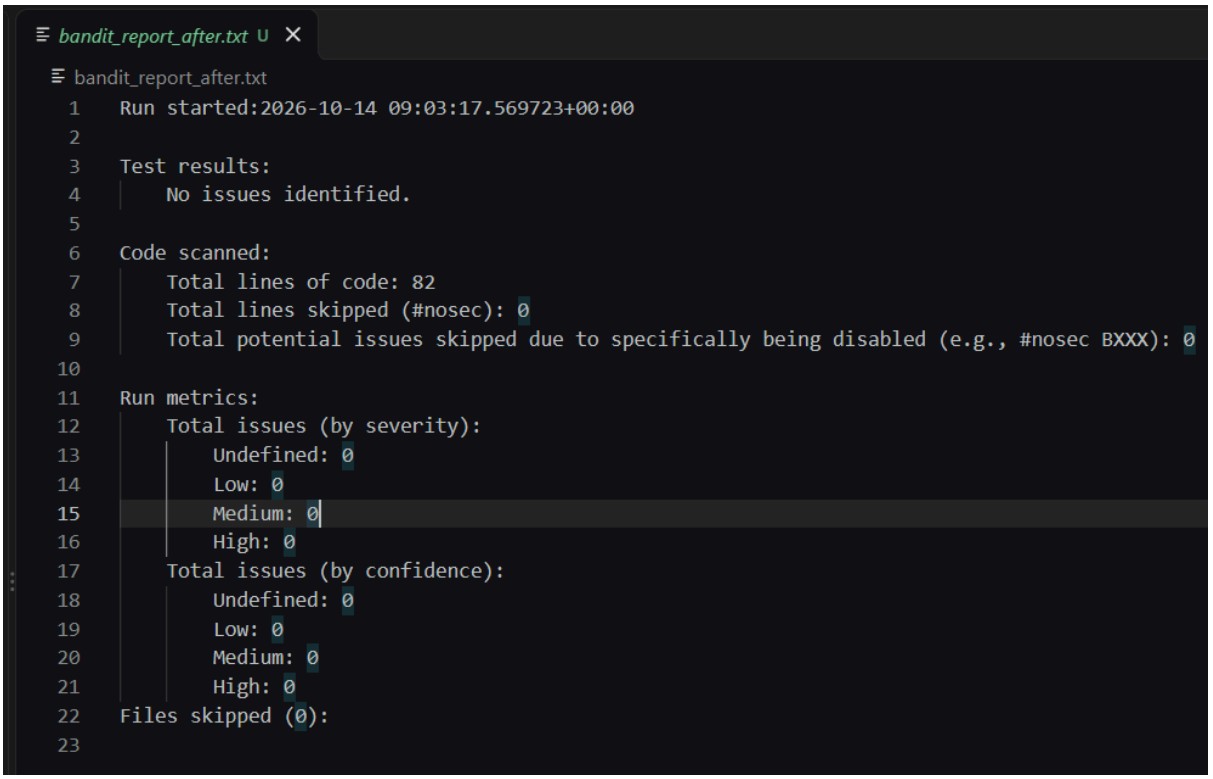
Issue	Type	Line(s)	Description	Fix Approach
Shell command injection (B605)	Security – High	64	os.system("echo ... " + filename) runs a shell command built from user input, so a crafted filename can run arbitrary commands.	Removed os.system and used print() to show the message.
Weak MD5 hash (B324)	Security – High	53	hashlib.md5 is cryptographically broken and can be attacked with collisions.	Replaced with hashlib.sha256.
Use of eval (B307 / W0123)	Security – Medium	48	eval(surge_expr) runs any string as Python code, which allows code injection.	Removed eval; the function now takes a numeric surge_multiplier and computes base_rate * surge_multiplier.
Hardcoded secret	Security – Medium	76, 85	The barrier key is hardcoded in the source and printed to the console.	Read the key from the GATE_BARRIER_KEY environment variable and print only whether it is set.
Mutable default argument (W0102)	Bug – Medium	15	entry_logs=[] is shared between calls, so logs pile up from earlier calls.	Default set to None, with a new list created inside the function.
Bare except: pass (W0702 / E722 / B110)	Bug – Medium	32-33	Every error, including real bugs, is silently swallowed.	Catch only KeyError, report the missing slot and return False.
Use of global (W0603)	Code Quality – Medium	12, 16	A global variable is changed from inside a function, so state is hard to track.	Replaced with a module-level parking_stats dict that is updated without global.
Insecure random generator (B311)	Security – Low	57	random.randint is predictable and should not be used for ticket PINs.	Used secrets.randbelow(9000) + 1000.

Issue	Type	Line(s)	Description	Fix Approach
Unused import / variables (W0611 / W0612 / F401 / F841)	Code Quality – Low	4, 37, 47, 69	math, overtime_fee, base_rate and the loop variables are never used.	Removed math, overtime_fee and s; base_rate is now used in the fee calculation.
Comparison to None (C0121 / E711)	Style – Low	38	Used == None instead of is None.	Changed to is None.
Simplifiable if / else after return (R1703 / R1705)	Code Quality – Low	40-43	An if/else that returns True/False is redundant.	Replaced with return not ...["is_occupied"].
Redefined outer name (W0621)	Code Quality – Low	67	The parameter parking_slots hides the global of the same name.	Renamed the parameter to slots.
Non snake_case names (C0103)	Style – Low	12, 15, 27, 36, 46, 52, 56, 60, 67	camelCase function names and a lowercase constant break PEP 8.	Renamed functions to snake_case (e.g. park_vehicle) and removed the global.
Missing docstrings (C0114 / C0116)	Style – Low	1, 15-75	No module docstring or function docstrings.	Added a docstring to the module and to every function.
Line too long / old-style formatting (C0301 / E501 / C0209)	Style – Low	7-9, 23	Lines exceed the length limit, and % formatting is used.	Split lines and use an f-string.
Missing final newline (C0304 / W292)	Style – Low	88	The file does not end with a newline.	Added a trailing newline and an if __name__ == "__main__": guard.

Pylint Report After Changes



Bandit Report After Changes



Flake8 Report After Changes

